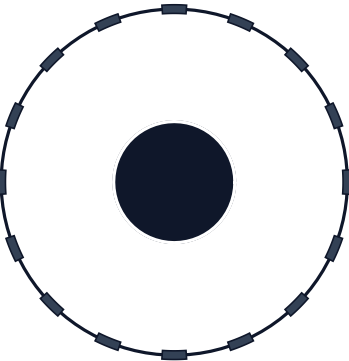
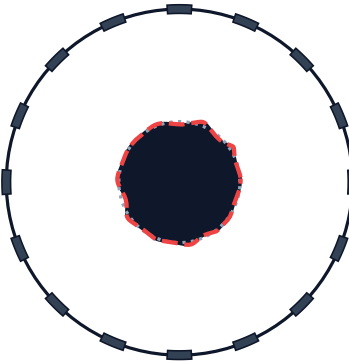


Surrogate-Assisted Levenberg--Marquardt Inversion (All 9 Targets)

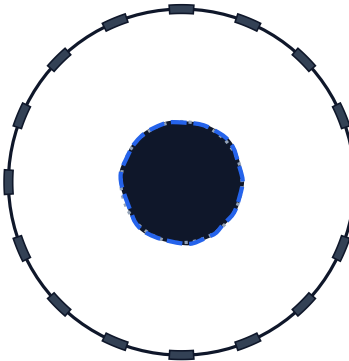
Target #1: Circle (Convex)
[Ground Truth]



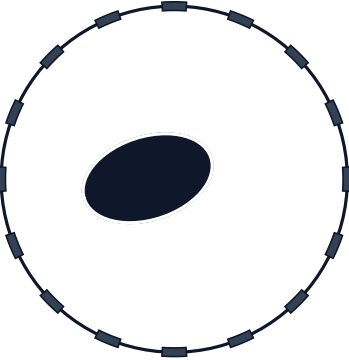
Deterministic Inversion
IoU = 0.9615 | RMS = 9.3 mm



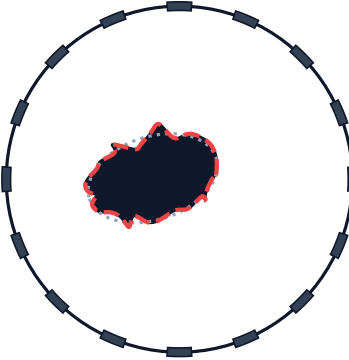
Ensemble Mean
IoU = 0.9765 | RMS = 6.0 mm



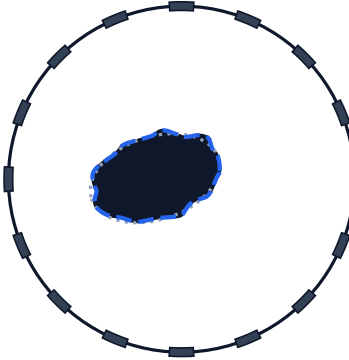
Target #2: Ellipse (Convex)
[Ground Truth]



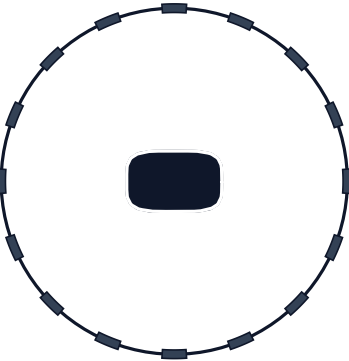
Deterministic Inversion
IoU = 0.9034 | RMS = 18.7 mm



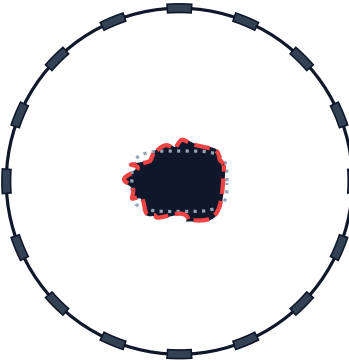
Ensemble Mean
IoU = 0.9377 | RMS = 12.5 mm



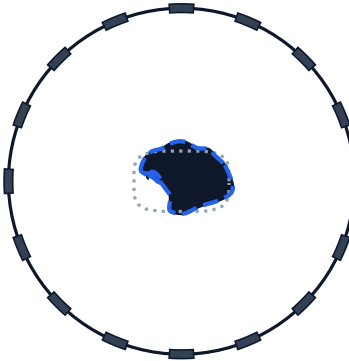
Target #3: Rectangle (Convex)
[Ground Truth]



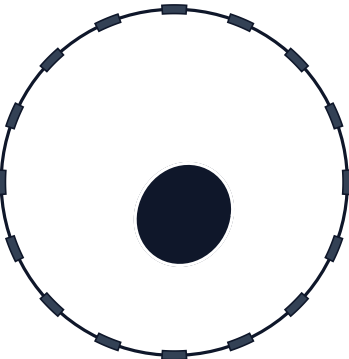
Deterministic Inversion
IoU = 0.7846 | RMS = 28.1 mm



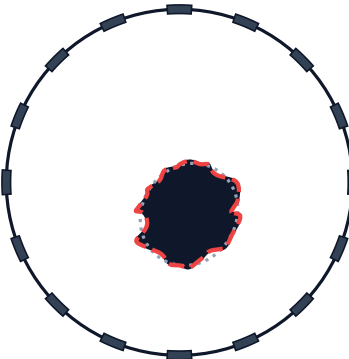
Ensemble Mean
IoU = 0.6610 | RMS = 70.0 mm



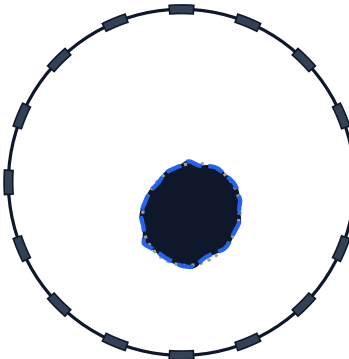
Target #4: Random Convex (Convex)
[Ground Truth]



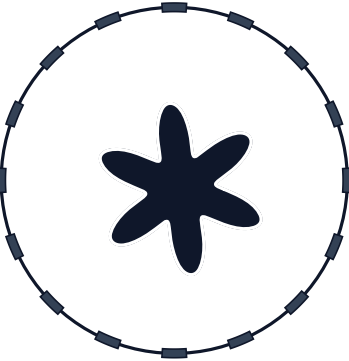
Deterministic Inversion
IoU = 0.9039 | RMS = 17.1 mm



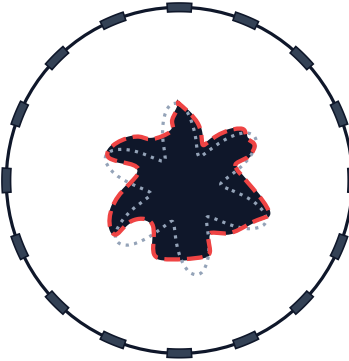
Ensemble Mean
IoU = 0.9557 | RMS = 8.4 mm



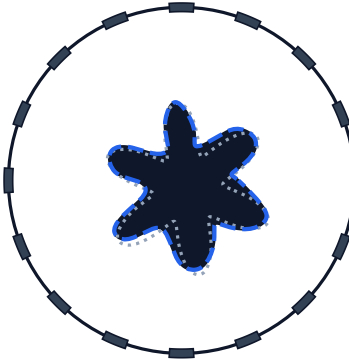
Target #5: Star (Concave)
[Ground Truth]



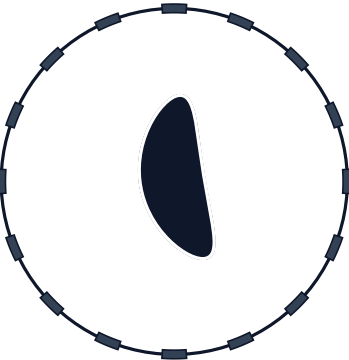
Deterministic Inversion
IoU = 0.6375 | RMS = 59.8 mm



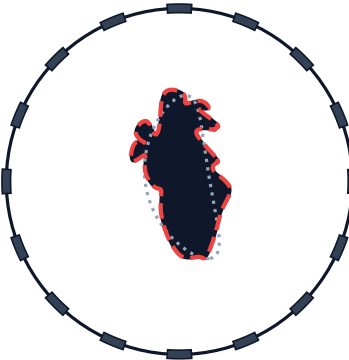
Ensemble Mean
IoU = 0.8243 | RMS = 28.6 mm



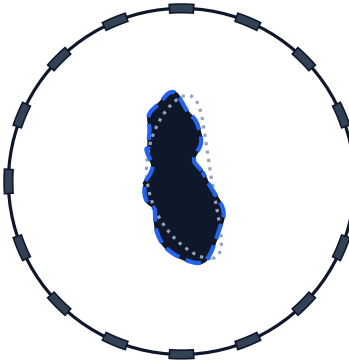
Target #6: Banana (Concave)
[Ground Truth]



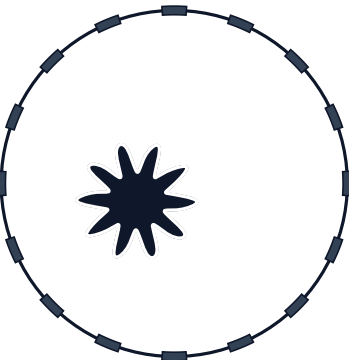
Deterministic Inversion
IoU = 0.7317 | RMS = 43.2 mm



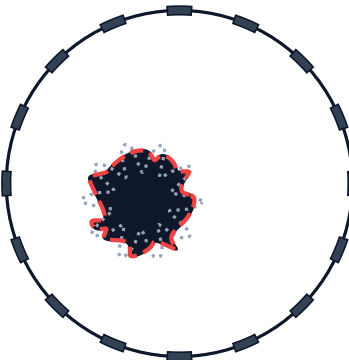
Ensemble Mean
IoU = 0.7405 | RMS = 41.6 mm



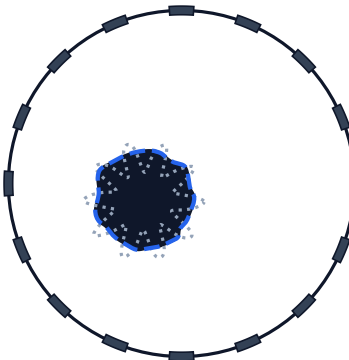
Target #7: Random Concave (Concave)
[Ground Truth]



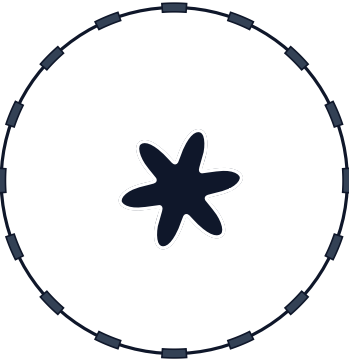
Deterministic Inversion
IoU = 0.6401 | RMS = 45.2 mm



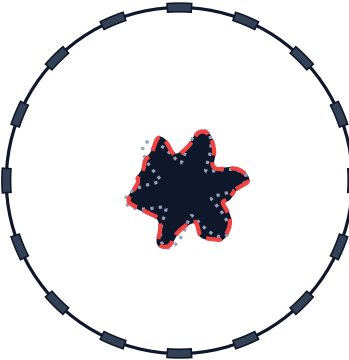
Ensemble Mean
IoU = 0.6650 | RMS = 49.8 mm



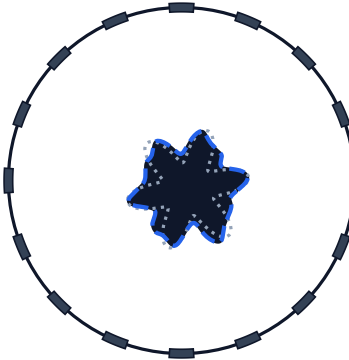
Target #8: Star (Concave)
[Ground Truth]



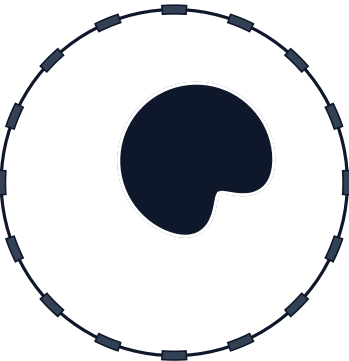
Deterministic Inversion
IoU = 0.7749 | RMS = 33.2 mm



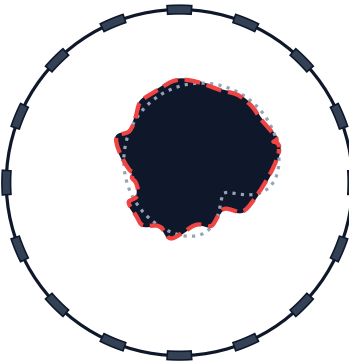
Ensemble Mean
IoU = 0.7581 | RMS = 36.6 mm



Target #9: Crescent (Concave)
[Ground Truth]



Deterministic Inversion
IoU = 0.8681 | RMS = 35.7 mm



Ensemble Mean
IoU = 0.8923 | RMS = 34.0 mm

